Search for heterotic line-bundle Standard Models

Summary of results, the algorithm, and directions beyond the heterotic string

Abstract. Much of what is known about string compactifications with the particle content of the supersymmetric Standard Model comes from computer scans. What a scan establishes depends on the status of each filter it applies. A filter that is not a genuine necessary condition can throw away real models, and one that is weaker than it could be lets spurious ones through. We revisit one of the most thoroughly scanned cases, the $E_8\times E_8$ heterotic string on favourable complete-intersection Calabi–Yau threefolds with a gauge bundle built from five line bundles, and rebuild the search so that a candidate is discarded only when a proof requires it. The main step is to replace the finite slope-stability test of earlier scans by its complete form, which asks whether a linear space meets a fixed cone. That question can be settled exactly on a partial model, so whole branches of the search are cut at once. The engine reproduces the published catalogs of the two benchmark manifolds model for model, and across all 68 manifolds it never misses a published model inside its search range. It also finds a stable GUT model that the catalog lacks, and eleven new three-family models on a quotient of the tetraquadric, each lying outside limits that the earlier standard-model scan set for itself. We then ask whether any of these models has a realistic superpotential, computing the leading couplings directly instead of inferring them from charges. For this quotient the answer is no: no model, published or new, combines a leading-order top-quark Yukawa coupling, Yukawa matrices that can reach full rank, and the absence of dangerous baryon- and lepton-number-violating operators. We end by describing how the same approach could be used for Type IIB orientifolds, flux vacua, F-theory and intersecting branes.

1. The problem

A bundle $V = \bigoplus_{a=1}^{5} O_X(k_a)$ on a favourable CICY X with $h = h^{1,1}(X)$ is an integer $h\times 5$ matrix $K = (k_1,\dots,k_5)$. For V to give a supersymmetric $SU(5)$ GUT with three chiral families that descends to an MSSM-like theory on X/Γ, with Γ acting freely, one needs $c_1(V) = 0$, $\mathrm{ind}\, V = -3|\Gamma|$, the Green–Schwarz condition $c_2(TX) - c_2(V) \geq 0$, and poly-stability. For line bundles poly-stability means that the five slopes $\mu_{k_a}(t) = d_{ijl}\, k_a^i t^j t^l$ vanish at a common point of the Kähler cone. The descent then requires an equivariant structure, a Wilson line and the Γ-representation on bundle cohomology. Earlier work turned this into scans over boxes $|k_a^i| \leq K$ [1, 2, 3], producing a catalog of 63,325 GUT models on the 68 favourable CICYs and a database of standard models with their superpotential selection rules.

Those scans leave three questions open: what a negative result proves, when several filters are finite truncations of conditions on infinitely many charge combinations; whether the downstairs spectrum can be computed from the group action rather than looked up; and what the superpotential does beyond charge conservation, which is necessary for a coupling but not sufficient.

2. The algorithm and what is new in it

The search is a depth-first traversal of sorted column prefixes $k_1 \preceq \cdots \preceq k_m$, $m \leq 4$. The fifth column is fixed by $\sum_a k_a = 0$, so every class of bundles is visited exactly once. Twelve rules prune the tree; each is a proven necessary condition, except one that is a declared restriction of the domain, and none is a heuristic.

A complete slope obstruction. Earlier scans reject K when some combination $\sum_a c_a k_a$ with $c \in \{-1,0,1\}^5$ has a one-signed matrix $M(k)_{jl} = d_{ijl}k^i$, since a one-signed quadratic form cannot vanish inside the Kähler cone. We show that the same test over all real c is a geometric statement: the span S of the charge vectors meets a fixed rational polyhedral cone $P = \{k : M(k) \geq 0\}$ away from $M = 0$. Three properties make this useful in a search. It is *monotone*, so a failure on a prefix prunes every completion. It is *decided exactly* by Gordan's alternative applied to the extreme rays of P projected onto $S^\perp$, whose dimension shrinks as the prefix grows; at the deepest and most numerous nodes it reduces to a sign check or an angular-gap check. And it is *strictly stronger* than the finite test: on the tetraquadric X_{7862} it removes one orbit that the finite test keeps at $K = 3$ and $K = 4$, and removing it is what brings the search into exact agreement with the published catalog.

Searches without a charge cutoff. A common slope zero forces a primitive integer vector p with all $p_i > 0$ and $\langle k_a, p\rangle = 0$ for every a, so each model lies in a slice $p^\perp$. When $(\sum_i p_i)^2 > (h-1)|p|^2$, the anomaly condition bounds every charge in the slice, and the slice can be searched over all integers. On X_{7862}, 425 of 469 such slices are proven empty for arbitrarily large charges. Separately, the greatest common divisor d_X of all line-bundle indices on X must divide $3|\Gamma|$; this excludes some pairs (X,Γ) for any charges, and both pairs it excludes among the 68 CICYs are indeed empty in the catalog.

An exact decision of stability. A bundle can pass every sign test while its five slopes have no simultaneous zero. When the charge matrix has rank $h-1$ we decide the question exactly: a lexicographic Gröbner basis, isolation of the real roots of each irreducible factor of the eliminant in rational intervals, interval arithmetic for positivity, and a check that every slope vanishes modulo the minimal polynomial. It returns either a real-algebraic Kähler class where V is poly-stable or a proof that none exists.

Selection rules as network flows. Whether the $U(1)^4$ charges allow a superpotential monomial after any number of singlet insertions is a monoid membership question. We show that it is a nonnegative-flow problem on a small directed graph. Total unimodularity makes the real relaxation exact, so the question is decided to all orders in the singlets, and a min-cost flow gives the lowest order at which a coupling can appear.

Cohomology and couplings from first principles. A nonzero $r\times r$ minor modulo a single prime certifies that a Koszul map has rank at least r for generic complex structure, with no good-reduction hypothesis. The Γ-representation on cohomology is computed from the group action through the Koszul resolution, Bott's theorem and equivariant map ranks. Leading-order Yukawa couplings are computed as cup products $H^1\times H^1\times H^1 \to H^3(X,O_X)$ [4] after splitting each class by Koszul level, which detects couplings that the charges allow but that vanish for every complex structure.

The pruning keeps the search small. On X_{7862}, from $K = 1$ to $K = 7$ the nominal domain grows by a factor 1.5×10^{11} while the number of complete bundles the search builds grows by a factor of about 700.

3. Results

Exact agreement with the published catalogs. The survivor sets coincide with the catalog model for model: all 10 and all 58 models on X_{7862} ($\Gamma = \mathbb{Z}_2$ and $\mathbb{Z}_2 \times \mathbb{Z}_2$) and all 93 on X_{7447}, whose counts we predicted before each run. Separate code sharing nothing with the search replays the complete $K = 1$ domains (4.3×10^7 and 3.5×10^9 states) with identical survivors.

All 68 favourable CICYs, and a model the catalog misses. In 271 runs covering every (geometry, $|\Gamma|$) key of the catalog ($h^{1,1} \leq 6$, $K \leq 3$) the engine never misses a published model inside its box (9,016 such models in the $K = 2$ runs alone). It also keeps 80 orbits the catalog does not list. The exact decision proves 53 of them unstable. Eighteen have charge matrices of rank below $h - 1$, where the decision does not apply, and are unstable on numerical evidence only. The remaining nine are a single model, seen on three CICY descriptions (5256, 5301, 5452) of one manifold. With $|\Gamma| = 4$ and charge vectors $(-2, 0, -1, 1, 0)$, $(0, -1, 1, -2, 1)$, $(0, 2, 1, 0, -1)$, $(1, -2, 1, 0, 0)$, $(1, 1, -2, 1, 0)$, it satisfies $c_1(V) = 0$, $\mathrm{ind}\, V = -12$ and the anomaly and Bogomolov conditions, and it is poly-stable with an exact witness: setting $t_5 = 1$, the coordinate t_4 is the root of an irreducible degree-12 integer polynomial isolated in $[642595/64736, 769226/77493]$. The catalog has 2,128 models on this key, 656 of them in the same charge box, and none with this model's intersection invariants. Its witness has $t_{\max}/t_{\min} \approx 231$, so a stability check confined to moderate Kähler ratios would not find it. It does not descend with individually equivariant line bundles under any of the four free $\mathbb{Z}_2 \times \mathbb{Z}_2$ actions, so it is a new GUT model, not a new standard model. Before trusting the verdict "unstable" we ran the decision on 548 published models, all poly-stable. The first version rejected 91 of them; this exposed two bugs, both fixed, and the check is now a regression test.

Downstairs spectra. Equivariant cohomology computed from the group action reproduces all 27,821 published character tables. From our own survivor set, 21 orbits on $X_{7862}/(\mathbb{Z}_2 \times \mathbb{Z}_2)$ have three chiral families, no charged chiral exotics and a Higgs doublet pair. All 10 published orbits are among them and 11 are new. Each of the 11 lies outside the published scan's own stated domain (its charge range, or its bound on vector-like pairs), which explains why it is missing.

Superpotential. The flow formulation reproduces all thirteen operator tables of the published standard-model database for its 2,012 complete models, once nine unstated conventions are read off from the data. It also shows that 954 of the 2,007 published down-type Yukawa tables are internally inconsistent: two families from the same line bundle receive different rows.

The physics verdict. We classified every configuration of the 21 orbits by explicit criteria: a nonzero top Yukawa at leading order, up- and down-type Yukawa matrices that can reach rank three, and dimension-five proton decay and dimension-four and bilinear R-parity violation forbidden to all orders. No orbit, published or new, meets them. Nine of the 11 new orbits fail on an exact obstruction, as do seven of the 10 published ones; two new orbits (A04, A06) remain undecided because the answer depends on higher-order coefficients and on the singlet vacuum. None of the new orbits has a leading-order top Yukawa, nine forbid dimension-five proton decay to all orders, and in one (A08) the charge-allowed coupling SLH_u vanishes cohomologically.

What is not claimed. No model is claimed to be the Standard Model. Cohomology is that of a generic member of each family. Supersymmetry breaking, moduli stabilisation and physical, canonically normalised Yukawa couplings are not computed. That survivor sets stop changing beyond $K = 4$ is recorded as an observation, not as a finiteness theorem. Table 1 lists the main results with their status.

result	status	domain
GUT catalogs of X_{7862} (10 and 58 models) and X_{7447} (93) reproduced	exact set equality	each full catalog
complete $K = 1$ domains replayed by independent code	exact	4.3×10^7, 3.5×10^9 states
425 of 469 slices of X_{7862} contain no model	theorem	all integer charges
$d_X \nmid 3\lvert\Gamma\rvert$ excludes (7761, $\lvert\Gamma\rvert = 2$) and (7447, $\lvert\Gamma\rvert = 5$)	theorem	all integer charges
no published model inside a search box is missed	exact	68 CICYs, 271 runs, $K \leq 3$
poly-stable GUT model on CICY 5256 absent from the catalog	exact (algebraic witness)	$\lvert\Gamma\rvert = 4$, $\lvert k\rvert \leq 2$
53 of 80 extra survivors unstable; 18 more	exact; numerical only	charge-matrix rank $h - 1$; below
27,821 published character tables reproduced	exact	whole database
21 downstairs orbits on $X_{7862}/(\mathbb{Z}_2 \times \mathbb{Z}_2)$, 11 new, each absence explained	exact, generic member	upstairs $K = 5$ survivors
13 operator tables reproduced; 954 of 2,007 Y_d tables inconsistent	exact	2,012 published models
no strong candidate; two orbits undecided	exact, perturbative	21 orbits
survivor set of X_{7862} unchanged for $K = 4, \ldots, 7$	observation, not a theorem	$\Gamma = \mathbb{Z}_2$

Table 1: Main results and the logical status of each.

Open problems in the heterotic setting. Whether X_{7862} carries finitely many models is open: the anomaly bound on the charges degenerates as the slope witness approaches the boundary of the Kähler cone, and we conjecture that the index condition keeps it away. Couplings with Koszul level sum two or more, which need a lift through the defining polynomial [5], are returned as unknown, and the 18 survivors of rank below $h - 1$ need an exact decision on positive-dimensional slope-zero loci. The survey should go to larger K, where further models missing from the catalog would appear, and to generalised CICYs [6] and hypersurfaces from the Kreuzer–Skarke list [7, 8], where the Kähler cone is no longer the positive orthant and the cone test must use the true one. An estimate of 10^{723} standard models on toric hypersurfaces [9] rules out box-by-box exhaustion there, so the charge-unbounded statements become the useful ones. Stability is decided at tree level; the one-loop correction [10] adds the string coupling as one more variable for the same exact methods.

4. Outlook: the same methods in other compactifications

What the engine exploits is a structure, not heterotic physics as such: integer data; a supersymmetry condition that asks whether a cone meets a linear space; a positive budget (anomaly or tadpole) that is spent along a branch of the search; and abelian selection rules. The same structure appears in most constructions that have produced the MSSM spectrum (Table 2).

construction	integer data	budget	supersymmetry condition	selection rules
heterotic line bundles (here)	charge matrix	$c_2(TX) - c_2(V) \geq 0$	slopes, quadratic in t; one loop [10]	$U(1)^4$ (flows)
IIB, magnetised D7-branes	divisors, fluxes	D7 and D3 tadpoles	FI terms, linear in t	anomalous $U(1)$s, E3
IIB closed-string fluxes	F_3, H_3	$N_{\text{flux}} \leq Q_{\text{D3}}$	$DW = 0$, polynomial near LCS	—
F-theory	G_4 flux	$\chi(Y)/24 - \frac{1}{2}G_4^2 \geq 0$	primitivity, linear in J	$U(1)$s, discrete
IIA, D6 on $T^6/\mathbb{Z}_2 \times \mathbb{Z}_2'$	wrapping numbers	RR tadpoles, K-theory	linear in complex-structure functions	anomalous $U(1)$s

Table 2: Where the ingredients of the search reappear (LCS: large complex structure).

Type IIB with magnetised D7-branes. A $U(1)$ stack on a divisor D_a with worldvolume flux F_a has, at leading order, Fayet–Iliopoulos term $\int_X J \wedge D_a \wedge F_a$, *linear* in the Kähler class. Whether all of them vanish at one point of the Kähler cone is then the question of Section 2, whether a rational subspace meets an open cone, but now necessary *and* sufficient, and settled by linear programming with an exact Farkas certificate either way. Chiral indices are integer intersection numbers, the D7 tadpole is a linear equality, Freed–Witten quantisation is a parity condition, and the D3 tadpole is a budget that worldvolume and closed-string fluxes share. The $SU(5)$ models of [11] show how restrictive these conditions are. A first experiment: exhaust the three-family $SU(5)$ D7 configurations on orientifolds of small-$h^{1,1}$ CICYs, and apply the flow rule to the anomalous $U(1)$s with E3-instanton charges added as generators, to decide which perturbatively forbidden couplings, the **10 10 5** top coupling among them, an instanton could supply.

Flux compactification and the shared tadpole. Fluxes fix the complex structure and the dilaton [12], and the Kähler moduli need non-perturbative effects [13] or α' corrections [14]. The tadpole conjecture [15] states that fixing all complex-structure moduli costs more D3 charge than is available when there are many moduli. Two computations follow directly. The perturbatively flat vacua of [16] are pairs of integer flux vectors such that a derived vector lies in the Kähler cone of the mirror and is orthogonal to one of the flux vectors, subject to integrality conditions and the tadpole bound: a cone meeting a subspace, and a monotone budget, which is what our search is built for. A complete, certified list of such vacua on small-$h^{2,1}$ examples would replace sampling with a theorem. The second asks how much of the D3 tadpole the cheapest certified three-family brane sector uses on a given orientifold, and whether the rest can fix the complex structure; this tests the conjecture example by example as a finite computation. Visible sectors on D3-branes at del Pezzo singularities [17], which meet the global data mainly through the tadpole, are a natural first target.

F-theory. In the $O(10^{15})$ compactifications with the exact chiral spectrum of the Standard Model [18], global consistency reduces to a criterion on the base; the vector-like spectrum and the Yukawa couplings must still be settled model by model. Proving vector-like exotics absent needs upper bounds on cohomology, that is lower bounds on ranks, which a minor modulo a prime certifies, and the flow rule applies unchanged to Mordell–Weil $U(1)$s. In G_4 searches the chiral indices are linear in the flux, and since the intersection form is positive definite on primitive $(2,2)$-classes the D3 tadpole confines the flux to an ellipsoid, which can be enumerated exactly, branch by branch.

Type IIA intersecting D6-branes. On $T^6/(\mathbb{Z}_2 \times \mathbb{Z}_2')$ the supersymmetry equality of each stack is linear in four positive functions of the complex-structure moduli [19, 20]. A common supersymmetric point therefore needs a positive vector orthogonal to the span of the stacks' wrapping data: the slice problem of Section 2, decided by the same Gordan test and monotone as stacks are added. All vacua of this orientifold have been counted exactly [21], and MSSM-like ones estimated at one in a billion [22]; restricting the exact count to the MSSM chiral spectrum and running the flow analysis on the anomalous $U(1)$s would give a certified list of three-family models together with the couplings each allows perturbatively.

In each case the work has the same shape: find the integer data, state every condition as exact, necessary or a declared restriction, and apply the monotone ones as early in the tree as possible. A search built this way settles what sampling cannot: that a model is absent, that a catalog is complete in a box, or that an allowed coupling vanishes.

[1] L. B. Anderson et al., Phys. Rev. D 84 (2011) 106005 [1106.4804].
[2] L. B. Anderson et al., JHEP 06 (2012) 113 [1202.1757].
[3] L. B. Anderson et al., JHEP 01 (2014) 047 [1307.4787].
[4] S. Blesneag et al., JHEP 01 (2016) 152 [1512.05322].
[5] S. Blesneag et al., JHEP 01 (2017) 119 [1607.03461].
[6] L. B. Anderson et al., arXiv:1507.03235.
[7] M. Kreuzer, H. Skarke, Adv. Theor. Math. Phys. 4 (2002) 1209 [hep-th/0002240].
[8] R. Altman et al., arXiv:1411.1418.
[9] A. Constantin et al., arXiv:1810.00444.
[10] R. Blumenhagen et al., JHEP 06 (2005) 020 [hep-th/0504232].
[11] R. Blumenhagen et al., Nucl. Phys. B 815 (2009) 1 [0811.2936].
[12] S. B. Giddings et al., Phys. Rev. D 66 (2002) 106006 [hep-th/0105097].
[13] S. Kachru et al., Phys. Rev. D 68 (2003) 046005 [hep-th/0301240].
[14] V. Balasubramanian et al., JHEP 03 (2005) 007 [hep-th/0502058].
[15] I. Bena et al., arXiv:2010.10519.
[16] M. Demirtas et al., arXiv:1912.10047.
[17] M. Cicoli et al., arXiv:1206.5237.
[18] M. Cvetič et al., Phys. Rev. Lett. 123 (2019) 101601 [1903.00009].
[19] M. Cvetič et al., Phys. Rev. Lett. 87 (2001) 201801 [hep-th/0107143].
[20] R. Blumenhagen et al., Ann. Rev. Nucl. Part. Sci. 55 (2005) 71 [hep-th/0502005].
[21] G. J. Loges, G. Shiu, arXiv:2206.03506.
[22] F. Gmeiner et al., JHEP 01 (2006) 004 [hep-th/0510170].